Budget Front Desk

An Agentplace Luna agent: failures, changes, and the rerun

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What I built and tested

A public Luna front desk for **Budget Heating, Cooling & Plumbing**, built from its public website. It answers price and service-area questions, triages emergencies, and routes to a booking page or phone. The company is not involved; the agent has no connection to its calendar or CRM.

The corpus retains **33 hand-written and 55 generated single-turn scenarios**, **18 scripts of three turns each**, and **20 sealed validation scenarios**. Tests include missing details, unknown prices, off-topic requests and pushback. The main single-turn comparison and multi-turn scripts run three trials each. A pass requires the specified facts and next step, with no forbidden claim. "All-three pass" means the scenario passed every trial.

Changes made, including the regressions

A0 is the baseline; A5 is the selected instruction set. A generic next-step instruction first produced "reply in chat" instead of the phone number. Requiring an **external action** repaired that, but introduced scheduling and off-task regressions. Asking when the office reopens then triggered a clock check that contradicted a Sunday-night emergency. That edit was reverted. A5 retained the external action and forbade clock-based contradictions of the customer's stated time.

Before and after on the shared development set

> **All-three passes: 26/53 → 43/53.** This is **in-sample**: the scenarios guided instruction and grader iteration.

Requirement group	Baseline	A5 instructions
Coverage	1/8	7/8
Diagnostic	6/10	10/10
Free Estimate	6/8	8/8
Trust	3/5	5/5
Emergency	4/8	6/8
Scheduling	3/9	4/9
Off-Task	2/3	2/3
Handoff	1/2	1/2

There are 18 gains and 1 loss (paired McNemar $p<0.001$). Most gains now state the $99 diagnostic or coverage rule. The lost sewage score reflects a pattern that misses keep-away wording, not established loss of safety advice. The external-next-step version reached 40/53. This sequential comparison is not a controlled ablation.

Validation and remaining work

The positive development result is not confirmed on the separate validation checks. These scores are automated rubric outcomes, not independently established customer success rates.

Multi-turn: a positive but uncertain difference

Across 18 scripts with three trials each, baseline passes **36/54** and A5 **41/54**; scripts passing all trials move from 9/18 to 12/18. There are 7 gains and 2 losses (nominal paired p=0.180, not statistically significant). Trials repeat scripts, so they are not 54 independent customers. Observed gains are concentrated in arithmetic and pushback, not a demonstrated effect of the instructions. One gain is the baseline's template-placeholder failure.

Sealed: no demonstrated advantage

The 20 scenarios, each three turns long, were authored by a different model. Scenario rubrics were frozen before replies. Before the render exclusion, totals are **17/20 baseline and 17/20 A5**; the subset not flagged as ambiguous is 12/13 versus 11/13.

A post-run mechanical amount-check fix, applied to both arms, moved A5 from 10/13 to 11/13; baseline was unchanged. Separately, cleaning excluded one A5 pass containing an appended render error, leaving **16/19 overall and 10/12 on the unambiguous subset**. Baseline had no exclusion. The small set does not establish equivalence or regression.

Remaining failures and grading limits

- **Scheduling: 4/9 all-three passes.** Some replies omit the required hours or Saturday surcharge. **Handoff: 1/2.** A multi-unit outage misses escalation.
- Six membership-script conversations fail in **both arms** on benefits omitted from v1; some also omit required pricing. These are mixed authoring and agent gaps.
- There are **21 logged grader corrections**: 20 from transcripts, one from a positive control. A correct overnight refusal still fails a wording-sensitive check; its recorded score is retained. Human calibration reused the same transcripts after a rule clarification, so it is not fresh-sample validation.

Build provenance and evidence

Single-turn files label the original project `jojuduewx7ud`. Validation compares the A5 rebuild `cgiefgaebmgh` with a **separate baseline rebuild**, `guikfrraxzk5`; build differences are an additional confound. The evaluated brief used 24 sourced fields. The current ledger has 25; its corrected v2 is not evaluated. Two builds also answered membership questions differently; the cause remains unknown.

All recorded runs are retained. The 20 sealed scenarios supplement, rather than replace, the larger development sets.

All test sets, runs, instruction history and findings: github.com/arjhinety/budget-front-desk. The published agent link is supplied separately.